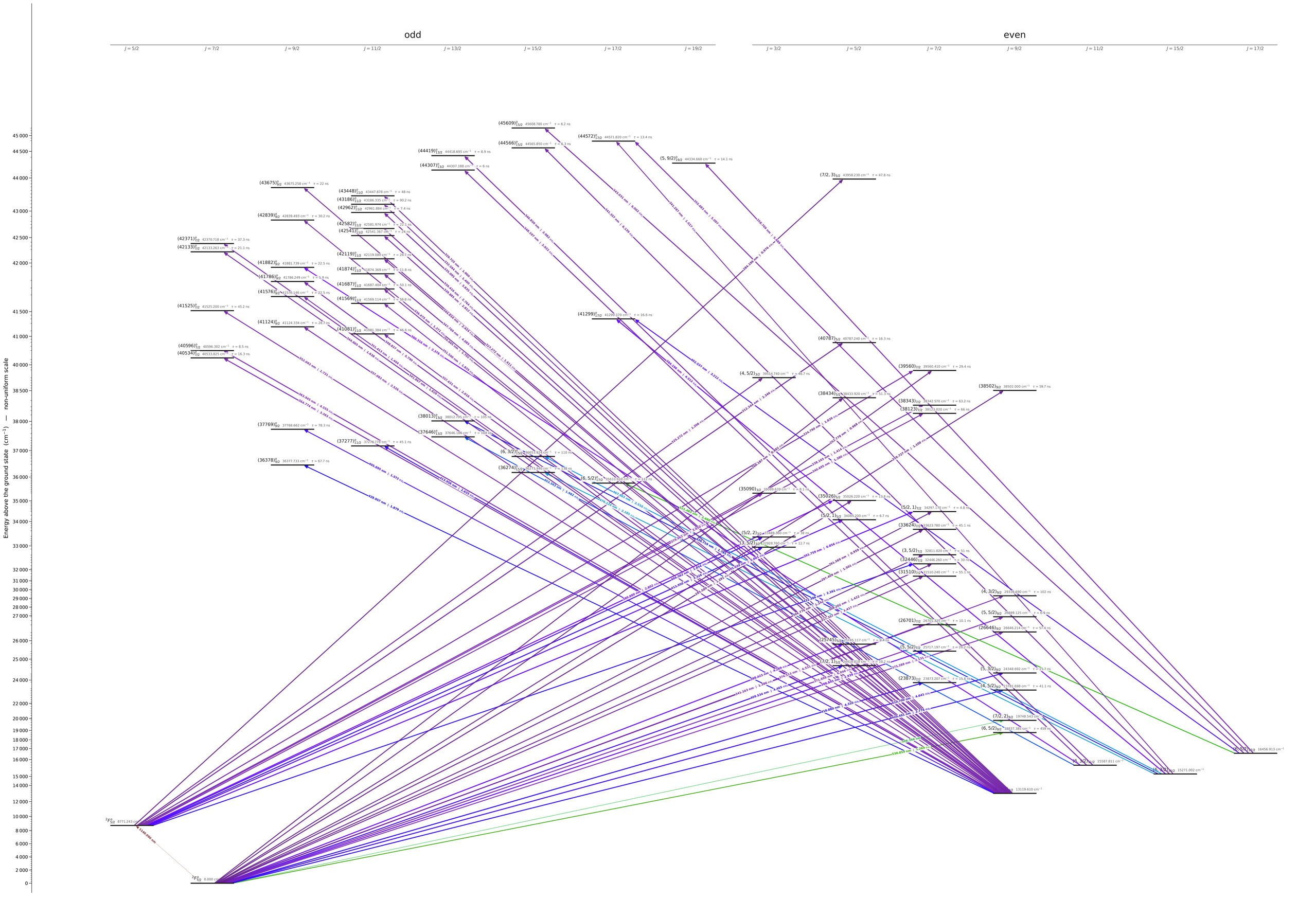
Thulium

Tm (Tm I). Strongest classified transitions below 2 µm (NIST): 72 levels, 73 lines.

Tm⁺ ionisation limit 49880.57 cm⁻¹ = 6.18440 eV (200.479 nm)

Arrow label: vacuum wavelength | reduced dipole matrix element |⟨J‖er‖J′⟩| in ea_0
Matrix elements are measured or NIST values; * = high-accuracy theory, ≈ = model calculation.
Arrow width grows with the Einstein coefficient A. Dotted: forbidden (clock, M1, E2). Dashed: Rydberg excitation, n = 70.
Level caption: energy (NIST, or a measurement of the last 20 years where one exists) and measured lifetime (where one exists).

Arrow colour = vacuum wavelength (nm)



Key transitions

Transition	λ vac (nm)	λ air (nm)	ν (THz)	|⟨J‖er‖J′⟩| (ea₀)	A (s⁻¹)	Γ/2π up. (MHz)	branch (%)	Use
$^2F^\circ_{7/2}$ – (5, 3/2)$_{9/2}$	410.6997	410.5838	729.9554	4.641	6.300e+07	10.14	100	first-stage
$^2F^\circ_{7/2}$ – (4, 5/2)$_{9/2}$	420.4914	420.3730	712.9574	2.776	2.100e+07	3.872	86.3	alternative blue cycling line
$^2F^\circ_{7/2}$ – (7/2, 2)$_{9/2}$	506.3665	506.2253	592.0464	–	–	–	–	third-stage deep narrow-line MOT
$^2F^\circ_{7/2}$ – (6, 5/2)$_{9/2}$	530.8592	530.7116	564.7306	1.269	2.180e+06	0.3467	100	second-stage narrow-line
$^2F^\circ_{7/2}$ – $^2F^\circ_{5/2}$	1140.0896	1139.7776	262.955164	–	7.000e+00	1.421e-06	78.4	inner-shell M1 optical clock transition

λ, ν from NIST level energies unless a measured frequency for this isotope is available. A: measured value or NIST.
Γ/2π: natural linewidth of the upper level, 1/(2πτ).

^{169}Tm hyperfine structure (I = 1/2)

Level	A (MHz)	B (MHz)	Shift of each F level from the centre of gravity (MHz)
$^2F^\circ_{7/2}$	-374.137661(3)	–	F=3: +841.81 F=4: -654.74
$^2F^\circ_{5/2}$	-704.648958	–	F=2: +1233.14 F=3: -880.81
(6, 5/2)$_{9/2}$	-422.11(3)	–	F=4: +1160.81 F=5: -949.75
(4, 5/2)$_{9/2}$	-317.32(16)	–	F=4: +872.63 F=5: -713.97
(23873)$_{7/2}$	352.75(18)	–	F=3: -793.69 F=4: +617.31
(5, 3/2)$_{9/2}$	-371.50(16)	–	F=4: +1021.62 F=5: -835.88
(7/2, 1)$_{5/2}$	-656.5(4)	–	F=2: +1148.82 F=3: -820.58

Measured A, B constants; sources in the data file.